

Name: Apurv Kumar Singh

SRN: PES1UG24CS077

Class: 5B

Use-Case Flow Specification

Use Case Name: Calculate Risk Score (Compute Borrower Credit Risk Rating)

1. Use Case Description This use case (included by "Submit Loan Request") describes the process by which the Micro-Lending platform evaluates a borrower's financial profile to calculate and assign a credit risk tier (Low, Medium, or High).

2. Actors

- **Primary Actor:** Borrower
- **Secondary Actor:** Peer Lender (views the assigned risk tier)

3. Preconditions

- The Borrower has successfully registered an account on the platform.
- The Borrower has initiated a new loan request and populated their complete financial profile, including income, active debts, and employment history.

4. Postconditions

- **Success:** The Borrower is assigned a verified Risk Tier (Low/Med/High), which is saved to the database and made visible to potential Peer Lenders.
- **Failure:** The risk assessment is aborted, no score is assigned, and the Borrower is prompted to complete their profile.

5. Main Success Scenario (Standard Flow)

1. The Borrower submits a loan request.
2. The system triggers the Risk Assessor Engine.
3. The system retrieves the borrower's submitted financial data: debt-to-income ratio, repayment history, and employment stability indicators.
4. The heuristic rules engine evaluates the retrieved indicators against the platform's risk baseline.
5. The system successfully calculates a numerical risk score.
6. The system maps the numerical score to the corresponding Risk Tier (Low, Med, or High).
7. The system updates the loan request status to "Assessed" and displays the tier to both the Borrower and Peer Lenders.

6. Alternate Flow 1 (Incomplete Financial Profile)

- *2a.* During step 3 of the Main Success Scenario, the system detects missing mandatory data (e.g., missing employment stability indicators).
- *2b.* The system halts the risk calculation process immediately.
- *2c.* The system generates an error notification: "Incomplete financial profile. Please update your employment history to proceed."
- *2d.* The credit approval fails, and the loan request remains in the "Draft" state until the Borrower updates the required fields.